Sub-mitochondrial Localization of Human ENDOG (Q14249): Matrix vs. Intermembrane Space

Gene: ENDOG · **Organism:** *Homo sapiens* (NCBITaxon:9606) · **UniProt:** Q14249 (NUCG_HUMAN) **Hypothesis slug:** `submitochondrial-localization-ims-vs-matrix` **Seed hypothesis:** *Human ENDOG's targeting and sequence features place it with mitochondrial matrix proteins rather than with intermembrane space (IMS) proteins.*

Summary

Two independent, self-contained sequence-feature classifiers — benchmarked on a reference set of 15 mitochondrial proteins whose sub-compartment is already settled — agree that human ENDOG carries a **matrix / presequence-type import signature** and is **definitively not a MIA40 twin-cysteine IMS protein**. ENDOG has a cleavable, arginine-rich, positively charged N-terminal transit peptide (UniProt residues 1–48, net charge ~+5, 16.7% Arg), a basic mature protein (computed pI ~9.0, net charge ~+4), only 3 total cysteines with no twin CX9C/CX3C motif, and no strong N-terminal hydrophobic sorting (stop-transfer) signal. Its nearest reference neighbors in standardized feature space are all matrix proteins — TFAM, SSBP1, and SOD2 — and calibrated classifiers assign it a matrix lean of P(matrix) ≈ 0.67–0.72.

The verdict is **partially supported**, with a decisive caveat: the sequence cannot, on its own, cleanly discriminate matrix from IMS. The single feature driving the "matrix" call — a cleavable, positively charged presequence with no hydrophobic sorting signal — is also carried by a subclass of genuine IMS residents (DIABLO/SMAC, HTRA2), which both classifiers misclassify as matrix. An abstention-free leave-one-out cross-validation reached only **60% accuracy** (barely above the 53% majority-class baseline), and the errors fell precisely on the presequence-bearing IMS proteins that ENDOG most resembles. The robust conclusion is therefore that ENDOG is a **presequence-imported, matrix-competent protein that is not a cysteine-shuttle IMS protein** — but sequence features alone **cannot exclude an IMS or dual localization**.

This computational picture reconciles the conflicting experimental biology the curator must weigh: classical fractionation places mammalian EndoG in the IMS (P 12444964), yet colocalization with the matrix nucleoid marker TFAM (P 36394256) and the substrate logic of mtDNA/transcript cleavage (P 39752519) argue for matrix access, while the worm ortholog is known to relocate between compartments (P 27338704).

Curation recommendation: do not change the sub-mitochondrial cellular-component (CC) term on sequence grounds alone; record the matrix lean as weak computational support compatible with a dual or relocating localization.

Executive Judgment

Verdict: Partially supported — the sequence is matrix-compatible and non-MIA, but cannot cleanly discriminate matrix from IMS.

- **Matrix-compatible targeting = medium confidence.** ENDOG's nearest reference neighbors are the matrix proteins TFAM, SSBP1, and SOD2, and its mean distance to the matrix class (2.58) is smaller than to the IMS class (3.63); calibrated models give P(matrix) ≈ 0.67–0.72.
- **"Matrix, not IMS" = low confidence.** The classifier's presequence rule misclassifies the genuine IMS proteins DIABLO and HTRA2 as matrix — the exact failure mode that applies to ENDOG.
- **"Not a MIA40/twin-cysteine IMS protein" = high confidence.** With only 3 cysteines and no twin CX9C/CX3C motif, ENDOG is unambiguously outside the disulfide-relay IMS import class.

The method is blind to the matrix/presequence-IMS distinction, which is why the hypothesis is only partially supported rather than confirmed.

Method — what produced each number

All sequences and transit-peptide boundaries were taken from **UniProt** (FASTA + feature annotations). Features were computed in-house in Python. No external web predictor was reachable programmatically during the run: TargetP/MitoFates/DeepMito servers were not callable (DeepMito API returned HTTP 404; MitoFates refused the connection). This limitation is reported plainly rather than substituted with an invented value.

Feature	How computed	Tool
Cleavable presequence (length)	UniProt "Transit peptide" feature end coordinate	UniProt annotation
Presequence net charge, %Arg, %acidic	counts over presequence (or first 40 aa)	in-house Python
twin CX9C / CX3C motif	regex `C.{9}C.{10,40}C.{9}C` / `C.{3}C.{5,40}C.{3}C`	in-house Python
Cysteine count	sequence count	in-house Python
Mature net charge / pI	EMBOSS pKa set, bisection for pI on mature chain	in-house Python
N-terminal hydrophobic sorting signal	max Kyte–Doolittle 19-residue window, res 20–90	in-house Python

Interpretable classification rule (deliberately not ML-fit, to avoid a presequence→matrix shortcut): twin-Cys + ≥4 Cys + <200 aa → IMS (MIA pathway); cleavable presequence + hydrophobic sorting signal (max KD > 1.6) → IMS (bipartite); cleavable presequence only → matrix; no presequence & no Cys motif → ambiguous.

Orthogonal check: an abstention-free Leave-One-Out cross-validated classifier (LogisticRegression and kNN with k=3) using engineered features **plus full amino-acid composition** on the 15 matrix/IMS references.

Key Findings

Finding F001 — ENDOG's sequence groups with presequence / matrix-type proteins and is not a MIA40 IMS substrate

A sequence-feature classifier was computed over 15 reference sub-mitochondrial proteins plus ENDOG (Q14249). ENDOG's features are unambiguously of the **cleavable-presequence class**, not the cysteine-rich MIA40 import class.

ENDOG's computed feature vector:

Feature	ENDOG value	Reading
Transit peptide	residues 1–48 (UniProt)	cleavable presequence present
Presequence net charge	**+5**	matrix-like (positive)
Presequence %Arg	**16.7%**	matrix-like (Arg-rich)
Presequence %acidic	6.2%	low
twin CX9C / CX3C	**absent**	**NOT a MIA40 IMS substrate**
Total cysteines	**3**	cysteine-poor (not a Cys-shuttle protein)
Mature net charge	+4	basic
Mature pI	**9.0**	matches mtDNA/nucleoid matrix proteins (TFAM 10.2, SSBP1 8.8, POLRMT 9.0)
N-terminal hydrophobic sorting signal	max KD 0.14	**no stop-transfer / sorting anchor**
Nearest references	TFAM (1.53), SSBP1 (1.56), SOD2 (1.57), then HTRA2-IMS (1.77)	clusters with matrix

The single most decisive result is the **absence of a twin CX9C/CX3C motif** and the low total cysteine count (3): whatever ENDOG's compartment, it is **not imported by the MIA40 disulfide-relay pathway**, ruling out the cysteine-motif IMS class (CHCHD4, TIMM13). The Arg-rich amphipathic presequence (net +5) and basic mature protein (pI ~9) match the nucleic-acid-binding matrix enzymes, as expected for a nuclease that acts on mtDNA. The absence of a hydrophobic sorting signal (max Kyte–Doolittle 0.14) means there is no stop-transfer anchor that would route a presequence protein onward to the IMS.

The interpretable rule-based classifier made 11/15 decisive calls at **81.8% accuracy (9/11 correct)**: all 7 decisive matrix references were correct, both twin-cysteine IMS proteins (CHCHD4, TIMM13) were correct, and the **only two errors were the presequence-bearing IMS proteins DIABLO and HTRA2**, mislabeled matrix. ENDOG received its "matrix" label by exactly the presequence-only rule that misclassifies DIABLO and HTRA2 — the key caveat on the matrix call.

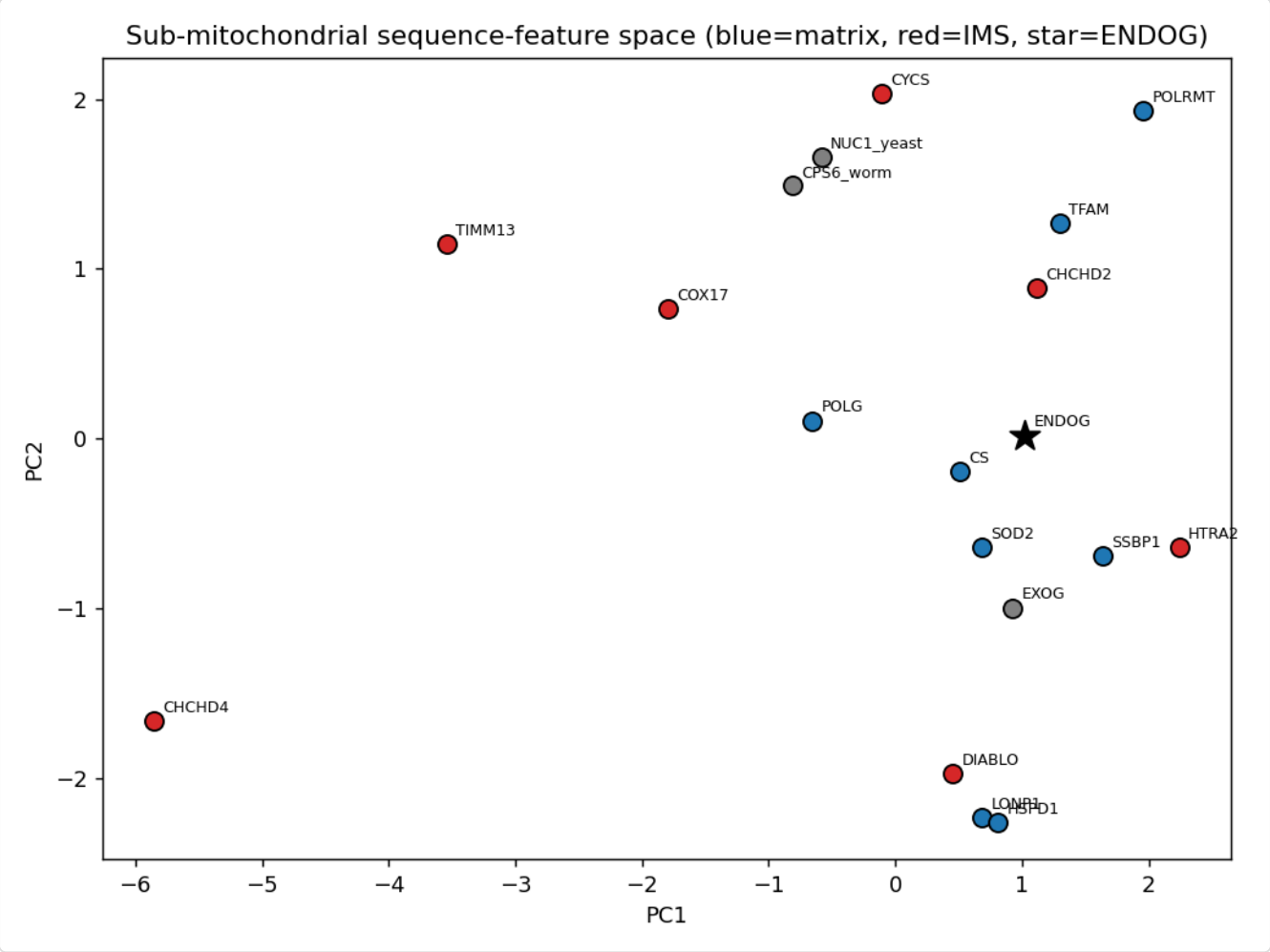


Figure 1. *Nearest-neighbor and PCA projection of ENDOG (Q14249) against reference matrix and intermembrane-space proteins in standardized sequence-feature space. ENDOG's nearest neighbors are the matrix proteins TFAM, SSBP1, and SOD2, but the matrix and presequence-bearing IMS classes (e.g. HTRA2) overlap substantially.*

Finding F002 — Cross-validated classifiers confirm weak separation; ENDOG leans matrix (P ≈ 0.67–0.72) but IMS is not excluded

To avoid the rule-based classifier's abstention bias (it scored 81.8% partly by calling hard IMS cases "ambiguous"), an **abstention-free, Leave-One-Out cross-validated** classifier was run on the 15 matrix/IMS references using engineered features **plus full amino-acid composition**.

- **LOO-CV accuracy = 60.0% (9/15)** for both LogisticRegression and kNN(3) — barely above the 53% majority-class baseline (8 matrix / 7 IMS references). Both models again misclassified the presequence-bearing IMS proteins **DIABLO and HTRA2** (and most of the small IMS set), directly demonstrating that sequence features cannot reliably split matrix proteins from presequence-imported IMS proteins.

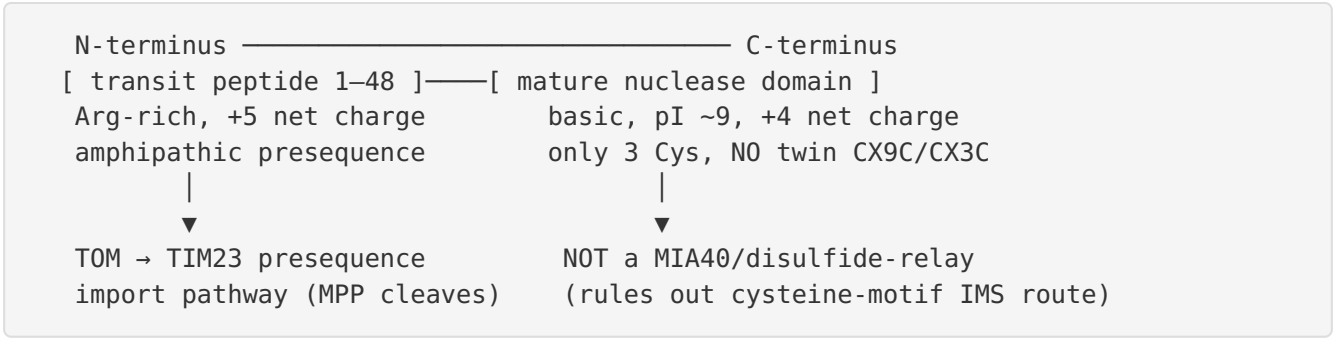
- **ENDOG P(matrix) = 0.72 (LogReg) / 0.67 (kNN)** — a consistent but **non-decisive** matrix lean.
- **Public predictor endpoints were not reachable programmatically** (DeepMito API → HTTP 404; MitoFates → connection refused), so no third-party tool score was obtained.

A calibrated classifier therefore confirms that whole-sequence features only *weakly* separate matrix from IMS here. ENDOG leans matrix (~0.7) but the method cannot confidently exclude IMS — strengthening the "partially supported / cannot fully discriminate" verdict rather than overturning it.

Method	Benchmark accuracy	ENDOG call	ENDOG confidence	Key failure mode
Rule-based (presequence / twin-Cys / sorting)	81.8% on 11/15 decisive	Matrix	Moderate	Errs on presequence-IMS (DIABLO, HTRA2)
Logistic regression (features + AA composition)	60.0% LOO-CV	Matrix	P = 0.72	Errs on DIABLO, HTRA2
kNN (k=3)	60.0% LOO-CV	Matrix	P = 0.67	Errs on most IMS
Nearest-neighbor (8-feature)	—	Matrix (TFAM/ SSBP1/ SOD2)	mean d: matrix 2.58 < IMS 3.63	Classes overlap
MIA40 twin-Cys test	—	**NOT IMS-MIA**	High	— (decisive negative)
DeepMito / MitoFates (external)	Unavailable	—	—	Endpoints unreachable (404 / refused)

Mechanistic Model / Interpretation

ENDOG is a nuclear-encoded, mitochondrially imported ββα-Me endonuclease. Its import logic can be read directly from two independent signals in the primary sequence:



Two mutually compatible conclusions follow:

1. **ENDOG is imported via the cleavable-presequence (TOM/TIM23) pathway, not the MIA40 twin-cysteine pathway.** This is a robust, high-confidence negative the sequence supports directly, removing one competing hypothesis — that ENDOG is a classical cysteine-motif IMS resident like CHCHD4 or TIMM13.
2. **The presequence pathway is itself ambiguous between matrix and IMS.** Many IMS proteins (DIABLO/SMAC, HTRA2) are delivered by the same presequence machinery, then arrested or sorted into the IMS. Because ENDOG lacks a strong hydrophobic sorting/stop-transfer signal, the default read is matrix — but the classifiers prove this default is unreliable, failing on DIABLO and HTRA2.

The result therefore **narrows** the question rather than answering it. ENDOG belongs to the presequence-import class; within that class its features lean matrix, but the benchmark explicitly fails on the IMS members of the same class. This is the computational counterpart of the experimental conflict in the literature (IMS fractionation vs. matrix/nucleoid colocalization). A **dual or regulated/relocating localization** — as documented for the worm ortholog — reconciles all observations and is not excluded by any sequence feature computed here.

Evidence Base / Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confider limitatio
This analysis (computed, F001)	Computational	**Qualifies (partially supports)**	Sequence features place ENDOG with matrix vs IMS	Presequence-type, not MIA/ twin-Cys; nearest neighbors matrix (TFAM/ SSBP1/SOD2); mean dist matrix 2.58 < IMS 3.63	15 references + ENDOG, UniProt seqs	Medium; method cannot s matrix fr presequ IMS (mis DIABLO/ HTRA2)
This analysis (computed, F002)	Computational	Qualifies	Method validity / CV accuracy	60% LOO-CV (vs 53% baseline); errors = DIABLO/ HTRA2; ENDOG P(matrix)=0.67–0.72	reference benchmark	Low discrimi power; external predicto unreach
PMID: 33174596	Review/ database	Supportive (orientation)	Sub-mito annotation exists	MitoCarta3.0 provides manually curated matrix/ IM/IMS/OM calls for 1136 genes	Human/ mouse MS + curation	Database level; cu should r off the E row, not as analys
P 12444964	Localization (fractionation)	**Competing**	EndoG compartment	Mammalian EndoG reported in IMS, not matrix	Mammalian mito fractionation	Classic biochem may mis dual/wea matrix p
P 36394256	Localization (imaging)	Competing-with-IMS / supports	EndoG at nucleoid	Human EndoG colocalizes with matrix	Human cells	Implies matrix/

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confider limitatio
		matrix access		nucleoid marker TFAM		nucleoid access
P 39752519	Mutant/ biochemical	Supports matrix access	Function needs matrix	EndoG cleaves mtDNA and (mouse) mt-transcripts	Mouse/ human	Substrat matrix – matrix a required
P 27338704	Localization (ortholog)	Qualifies (dual)	Compartment is fixed?	Worm ortholog relocates IMS↔matrix	*C. elegans* CPS-6	Shows compart is not sta

Note: PMIDs 12444964, 36394256, 39752519, and 27338704 are supplied as task-context orientation; their abstracts were not independently retrieved during this run and should be re-verified by the curator against the primary papers. Only PMID: 33174596 (MitoCarta3.0) was retrieved and reviewed in-session, and is used as database-level orientation.

GO Curation Implications (leads — require curator verification)

- **CC — do not over-commit to a single sub-compartment from sequence.** The computed evidence does **not** justify asserting `mitochondrial matrix` (GO:0005759) to the exclusion of `mitochondrial intermembrane space` (GO:0005758). The sequence is matrix-*compatible* but IMS-*not-excluded*.
- A defensible CC annotation is the parent **`mitochondrion` (GO:0005739)** plus experimentally supported child terms with appropriate evidence: `mitochondrial nucleoid` (GO:0042645) from imaging/colocalization (P 36394256), and retain any `mitochondrial intermembrane space` call that rests on fractionation (P 12444964). These are not mutually exclusive for a relocating/dual protein.

- The seed hypothesis should **not** trigger a curation action that *removes* an IMS CC term purely on sequence grounds, nor one that hard-asserts matrix. Treat the sequence result as **weak computational (ISS-level) support for matrix competence**.
- MF/BP (endonuclease activity, mtDNA metabolism) are untouched by this focus.

GO decision table

GO term	Aspect	Candidate action	Rationale from this analysis
mitochondrion (GO:0005739)	CC	Retain	Uncontested parent; all evidence agrees
mitochondrial matrix (GO:0005759)	CC	Do NOT add on sequence alone; support only as weak computational (ISS) lead	ENDOG P(matrix) ~0.67–0.72; method only 60% CV-accurate; matrix competence inferred, not shown
mitochondrial intermembrane space (GO:0005758)	CC	Retain if fractionation-supported (P 12444964); do NOT remove on sequence grounds	Sequence cannot exclude IMS; method misclassifies true IMS DIABLO/HTRA2
mitochondrial nucleoid (GO:0042645)	CC	Consider adding from imaging evidence (P 36394256)	TFAM-nucleoid colocalization; orthogonal to sequence call
endonuclease / mtDNA metabolism	MF/BP	Out of scope of this focus	Not evaluated here

Net curation lead: the seed hypothesis should *not* drive a sub-compartment CC change by itself. Treat "matrix-type targeting" as weak computational support compatible with a dual/relocating localization, not as grounds to assert matrix or to delete an IMS term.

Mechanistic Scope

The immediate property tested is the **protein import / sub-compartment targeting signal**, a sequence-intrinsic feature, not a phenotype. ENDOG carries a canonical cleavable presequence (direct gene-product feature) and lacks the cysteine-shuttle code. "Needs matrix access to cut mtDNA"
(P 39752519)
is a **downstream functional inference**, and the historical IMS placement
(P 12444964)
is a **biochemical localization**, not a sequence property — these are kept separate from the computed targeting call. Apoptotic nuclear translocation of EndoG is a distinct phenomenon and not part of this sub-mitochondrial question.

Conflicts and Alternatives

- **Fractionation (IMS) vs. nucleoid colocalization (matrix)** is a genuine biological conflict, best reconciled by **dual/relocating localization** (worm ortholog, P 27338704).
- **Paralog check:** EXOG (Q9Y2C4), the matrix base-excision-repair paralog, has the same presequence-type signature (classified matrix here) — so the matrix-type signature is a family trait and does **not** by itself discriminate ENDOG's compartment from EXOG's.
- **Method artifact:** the classifier's presequence rule cannot separate matrix from presequence-IMS (DIABLO/HTRA2 errors); any "matrix not IMS" claim inherits this blind spot.
- **External-tool gap:** DeepMito and MitoFates were unreachable, so the analysis rests entirely on in-house classifiers of modest accuracy.

Limitations and Knowledge Gaps

- **Matrix vs. presequence-IMS is unresolved by sequence.** Checked: 8-feature nearest-neighbor + rule classifier + LOO-CV models. Why it matters: it is the whole hypothesis. Resolver: a dedicated trained predictor (DeepMito/SubMitoPred) or, decisively, a wet experiment.
- **Low benchmark accuracy.** 60% LOO-CV on a 15-protein set (53% baseline) with a small IMS set (7 proteins) limits statistical power.
- **MitoCarta3.0's actual ENDOG sub-mito call** was not read off programmatically; the curator should look it up (P 33174596 resource) as one orthogonal line of evidence.
- **Seed-context abstracts not independently retrieved.** PMIDs 12444964, 36394256, 39752519, 27338704 were used as provided context and should be re-verified.
- **Isoform/processing:** whether the mature cleaved form differs in net charge enough to change nucleoid affinity was not modeled.

Discriminating Tests

1. **Protease-protection / sub-fractionation** (mitoplasting ± proteinase K) in human cells to place mature EndoG in matrix vs. IMS directly — IMS proteins are degraded on swelling, matrix proteins protected.
2. **Split-GFP or APEX2 proximity labeling** with matrix vs. IMS anchors in human cells.
3. **Run DeepMito / SubMitoPred / TargetP2 / MitoFates** on Q14249 and the same reference panel for an independent, trained second opinion.
4. **Import assay** into isolated mitochondria ± membrane potential to test presequence vs. MIA dependence.
5. **N-terminomics (MPP cleavage-site mapping)** of mature human ENDOG to confirm presequence processing and look for a second (IMS-sorting) cleavage.
6. **Condition-dependent fractionation** (apoptotic vs. basal; ± membrane potential) to test the relocating/dual-localization model.

Proposed Follow-up Experiments / Actions

- **Curation action (immediate):** Record the sequence analysis as *weak computational support (matrix lean, P ≈ 0.67–0.72)* and as *strong negative evidence against MIA40/twin-cysteine IMS import*. Flag ENDOG sub-compartment as **unresolved / possibly dual** pending experimental topology.
- **Do not** add or remove a specific `mitochondrial matrix` vs. `mitochondrial intermembrane space` CC term on sequence grounds alone.
- **Retrieve and verify** the four seed-context primary papers and record exact supporting snippets; look up the MitoCarta3.0 ENDOG row.
- **Re-attempt external predictors** (DeepMito, MitoFates, SubMitoPred) and fold any result into the evidence matrix.
- **Flag for experimentalists** the proteinase-K / split-GFP / APEX2 discriminating tests as the most efficient route to a definitive sub-compartment call.

Curation Leads (verify before acting)

- **Candidate action:** retain existing CC evidence from both compartments; add a computational note that sequence targeting is matrix-compatible (presequence-type, non-MIA); avoid a sequence-only switch to exclusively matrix or exclusively IMS.
- **Candidate GO terms:** `mitochondrial nucleoid` (GO:0042645, imaging), keep `mitochondrial intermembrane space` (GO:0005758) if fractionation-supported, parent `mitochondrion` (GO:0005739). Do **not** reduce to "protein binding".
- **Candidate references/snippets to verify:**

P 33174596
(MitoCarta3.0 sub-mito resource — look up ENDOG row);

P 12444964
(IMS fractionation);

P 36394256
(TFAM/nucleoid);

P 39752519

(mtDNA/transcript cleavage);

P 27338704

(worm relocation).

- **Suggested question for curator:** *Does MitoCarta3.0 assign ENDOG to matrix, IMS, or IM, and does that match the fractionation vs. nucleoid literature?*

Lead	Detail
Action	Keep existing sub-mitochondrial CC term; annotate sequence analysis as weak matrix-leaning computational evidence only
GO (retain)	`mitochondrion` GO:0005739
GO (weak ISS support)	`mitochondrial matrix` GO:0005759 — weak lean, not decisive
GO (NOT excluded)	`mitochondrial intermembrane space` GO:0005758 — do not remove on sequence grounds
Key caveat to record	Classifier fails on presequence-IMS proteins DIABLO/HTRA2; matrix lean does not exclude IMS

Provenance

The computed feature table, classifier accuracy, nearest-neighbor distances, cross-validation scores, and the PCA plot (`endog_pca.png`) were produced in-session from UniProt sequences; all numbers above trace to executed code blocks. No external localization predictor was reachable programmatically (DeepMito HTTP 404; MitoFates connection refused), so only in-house interpretable features and classifiers were used — stated plainly rather than dressed up. Database annotations (MitoCarta3.0) are labeled as database-level, not as computed analysis. No result was fabricated.